

MPC-PiNNs technique applied to Quadrotors

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1 Introduction

outline of the work

unicycle approach

uav approach

Model Predictive Control (MPC) has emerged as a powerful control strategy in robotics due to its ability to handle constraints and optimize system behavior over a prediction horizon. When combined with neural networks, which can learn complex system dynamics, it creates a promising framework for controlling nonlinear systems.

This report presents the implementation and analysis of a Neural Model Predictive Control (NMPC) scheme applied to a quadrotor. The unicycle serves as an excellent test case due to its nonlinear dynamics, making it both challenging and representative of real-world robotic systems. Besides, we followed a bottom-up approach, testing the pipeline first on a unicycle, which still has a nonlinear dynamics, but is easier to treat due a simpler model and faster prototyping possibilities.

Our approach integrates deep learning techniques to approximate the quadrotor's dynamics, which are then utilized within an MPC framework to generate optimal control inputs. We explore the effectiveness of this method in trajectory tracking tasks, comparing it with traditional control approaches.

The following sections detail the theoretical foundations, implementation specifics, and experimental results of our neural MPC system. We also discuss the challenges encountered and potential improvements for future work.

2 Model Predictive Control

uav simplified model

uav complete model

mpc structure

Model Predictive Control (MPC), it is a control technique used to solve an optimization problem at each sampling instant. The controller predicts the future behavior of the system using a mathematical model and determines a sequence of control inputs that minimize a predefined cost function while satisfying state and input constraints.

$$\min_{u(0), u(1), \dots, u(N-1)} \sum_{k=0}^{N-1} \ell(x(k), u(k)) + \ell_f(x(N)) \quad (1)$$

$$\text{s.t. } x(k+1) = f(x(k), u(k)), \quad k = 0, \dots, N-1, \quad (2)$$

$$x(k) \in \mathcal{X}, \quad u(k) \in \mathcal{U}, \quad k = 0, \dots, N-1, \quad (3)$$

$$x(0) = x_0 \quad (4)$$

One of the main advantages of this control approach is the ability to handle multivariable interactions and constraints. However, solving the above optimization problem in real time can be computationally challenging, especially for large-scale or highly nonlinear systems. This is one reason why neural networks have been employed to either approximate the system model or enhance the optimization process. In our case, we have used Neural Networks to predict the next state after the application of the control input given by the MPC.

3 Physical Informed Neural Network

Physical Informed Neural Networks (PiNNs) are a recent approach that exploit physical laws inside the learning process of a Neural Network. As previously stated, we have used a bottom-up approach by first testing this approach to a Unicycle. This robot still has a nonlinear dynamics but it's way simpler than the quadrotor one. Its configuration vector is composed by the cartesian position (x, y) and its orientation angle θ .

$$\begin{cases} \dot{x} = v \cos(\theta) \\ \dot{y} = v \sin(\theta) \\ \dot{\theta} = w \end{cases} \quad (5)$$

$$\mathbf{R}_{\text{RPY}} = \begin{bmatrix} \cos \psi \cos \theta & \cos \psi \sin \theta \sin \phi - \sin \psi \cos \phi & \cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi \\ \sin \psi \cos \theta & \sin \psi \sin \theta \sin \phi + \cos \psi \cos \phi & \sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi \\ -\sin \theta & \cos \theta \sin \phi & \cos \theta \cos \phi \end{bmatrix} \quad (6)$$

$$\dot{\mathbf{v}} = \begin{bmatrix} 0.0 \\ 0.0 \\ -g \end{bmatrix} + \mathbf{R}_{\text{RPY}} \cdot \begin{bmatrix} 0.0 \\ 0.0 \\ \frac{T}{m} \end{bmatrix} \quad (7)$$

In our implementation, we have used a simplified dynamic model for the quadrotor instead of the simplified second order system.

$$\begin{cases} \ddot{x} = -(\cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi) \frac{T}{m} \\ \ddot{y} = -(\sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi) \frac{T}{m} \\ \ddot{z} = -(\cos \theta \cos \phi) \frac{T}{m} + g \\ \ddot{\phi} = \frac{\tau_{\phi}}{I_x} \\ \ddot{\theta} = \frac{\tau_{\theta}}{I_y} \\ \ddot{\psi} = \frac{\tau_{\psi}}{I_z} \end{cases} \quad (8)$$

$$\begin{cases} \dot{x} = v_x \\ \dot{y} = v_y \\ \dot{z} = v_z \\ \dot{v}_x = (\cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi) \frac{T}{m} \\ \dot{v}_y = (\sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi) \frac{T}{m} \\ \dot{v}_z = (\cos \theta \cos \phi) \frac{T}{m} - g \\ \dot{\phi} = \\ \dot{\theta} = \\ \dot{\psi} = \end{cases} \quad (9)$$

In our implementation, we have used a PiNNs to predict the next state of the system, given the control input obtained from the Model Predictive Control.

losses

pipeline

image pinn

4 Results

what we have obtained

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5 Conclusions